

Tech Hardware - Asia Pacific

Optical transceiver expert call: leaders' positions to stay solid, more new techs

Industry Overview

Expert sees solid optical transceiver demand into 2030

We hosted a call with optical transceiver expert Ms. Qi Caijie on 20 April. Any views or information shared reflect Ms. Qi's opinions and not ours. Key takeaways: (1) 800G/1.6T demand is likely to reach 50-60mn /30mn units in 2026, while the industry meets 70-80% /50-60% of the demand this year; (2) SiPh penetration seen at 50%/60% for 800G/1.6T in 2026, and capacity and cost vs. EML will be keys to look at SiPh penetration going forward; (3) into 2027, 800G demand to remain relatively flattish and 1.6T demand to grow by at least 50% – meanwhile, pluggable to remain as the mainstream into 2030; and (4) NPO/XPO and 3.2T pluggable to see materialization into 2027-28.

Component shortage could ease from 3Q26

Ms. Qi noted tight supply across EML/DSP and even faraday rotator. Leading suppliers could see overall better procurement capability, thanks to larger scale and better resources to secure supply via prepayment. The expert expects supply situation to ease heading into 3Q, driving tier-1 suppliers' revenue QoQ. Among China laser suppliers, Ms. Qi believes Yuanjie and Source Photonics have the capability to supply EML lasers to US CSPs. Yuanjie also has edges like a close relationship with Innolight and a US facility, while future progress depends on its capability to mass produce 200G EML lasers.

Leaders to remain in front line despite new entrants

Despite more entrants in optical transceiver, Ms. Qi believes leaders like Innolight, Coherent and Eoptolink will continue to dominate. Although CSPs are testing/certifying more small suppliers due to tight supply, they are likely to continue limiting the supplier base to 3-4 through bidding, with tier-1 suppliers holding 50%+ share and the remainder split among the rest.

NPO/XPO volume to take off from 2027, 3.2T from 2028

Ms. Qi expects NPO/XPO shipment to start from 2027. Google plays a key role in driving NPO deployment, and other CSPs like Meta could follow suit. Whether the industry can meet 10 mn units' NPO volume in 2027 or not depends on certification (the expert also mentioned about 1.6T pluggable facing some delays in 2025). Meanwhile, suppliers are also looking for new designs to replace external lasers of NPO. For 3.2T NPO, Ms. Qi expects single-lane pricing to be 70-80% that of the current 1.6T transceivers, while 12.8T XPO to have a higher ASP than eight 1.6T transceivers. For 3.2T pluggable, Ms. Qi noted the industry is still under R&D and key bottlenecks remain at upstream components, including 400G EML and TFLN. Innolight still leads the way in R&D (followed by Coherent), and suppliers like Innolight are trying all designs, including TFLN-based, SiPh+TFLM, and EML-based ones. If R&D can be finished by end-2026, we could see volume take-off into 2028.

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Acronym

CPO: Co-packed Optics

CSP: Cloud Service Provider

DSP: Digital Signal Processor

EML: Electro-absorption-modulated Laser

NPO: Near-packaged Optics

SiPh: Silicon Photonics

TFLN: Thin Film Lithium Niobate

XPO: Extra-dence Pluggable Optics

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